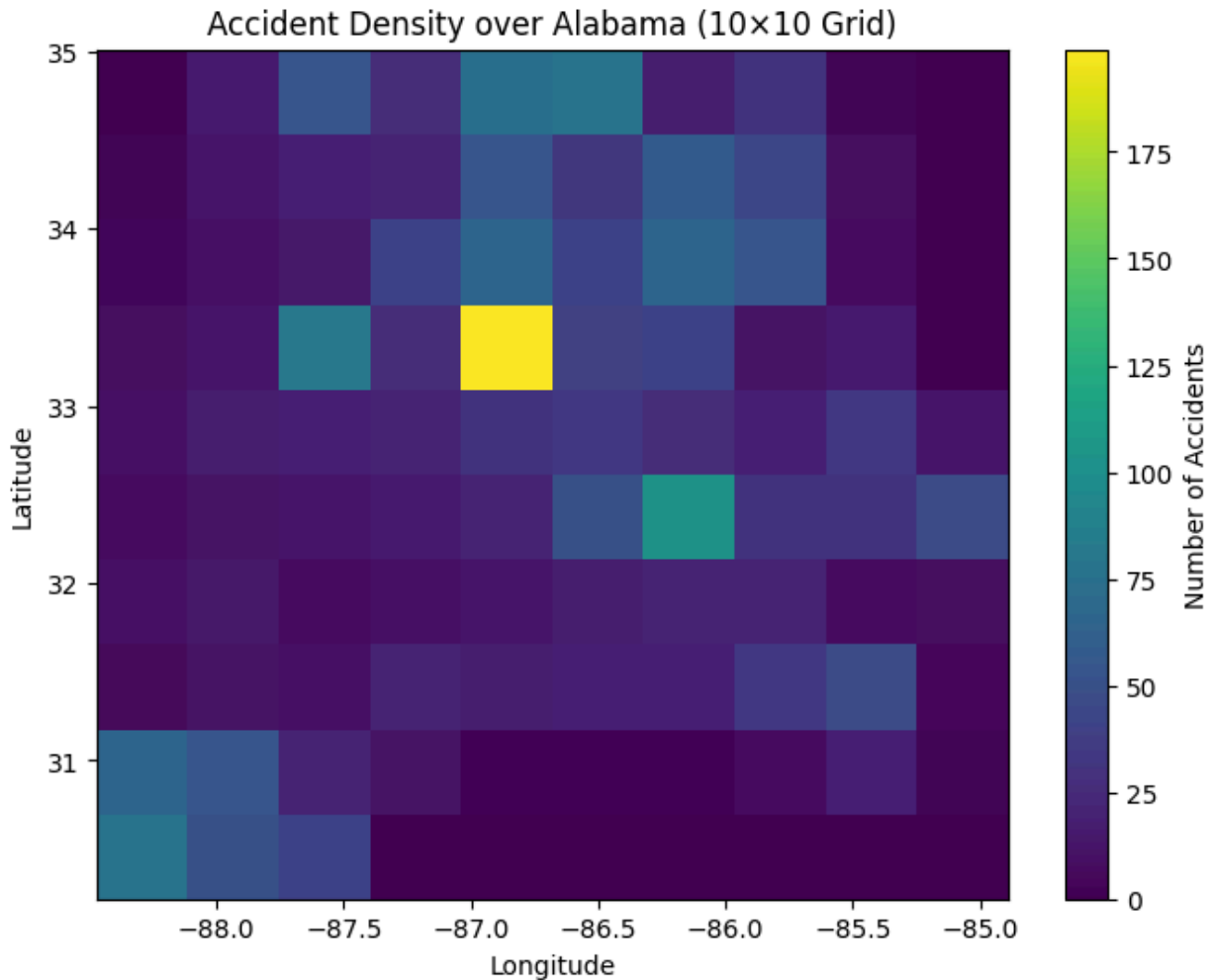


Traffic Accidents Alert System - Progress Report

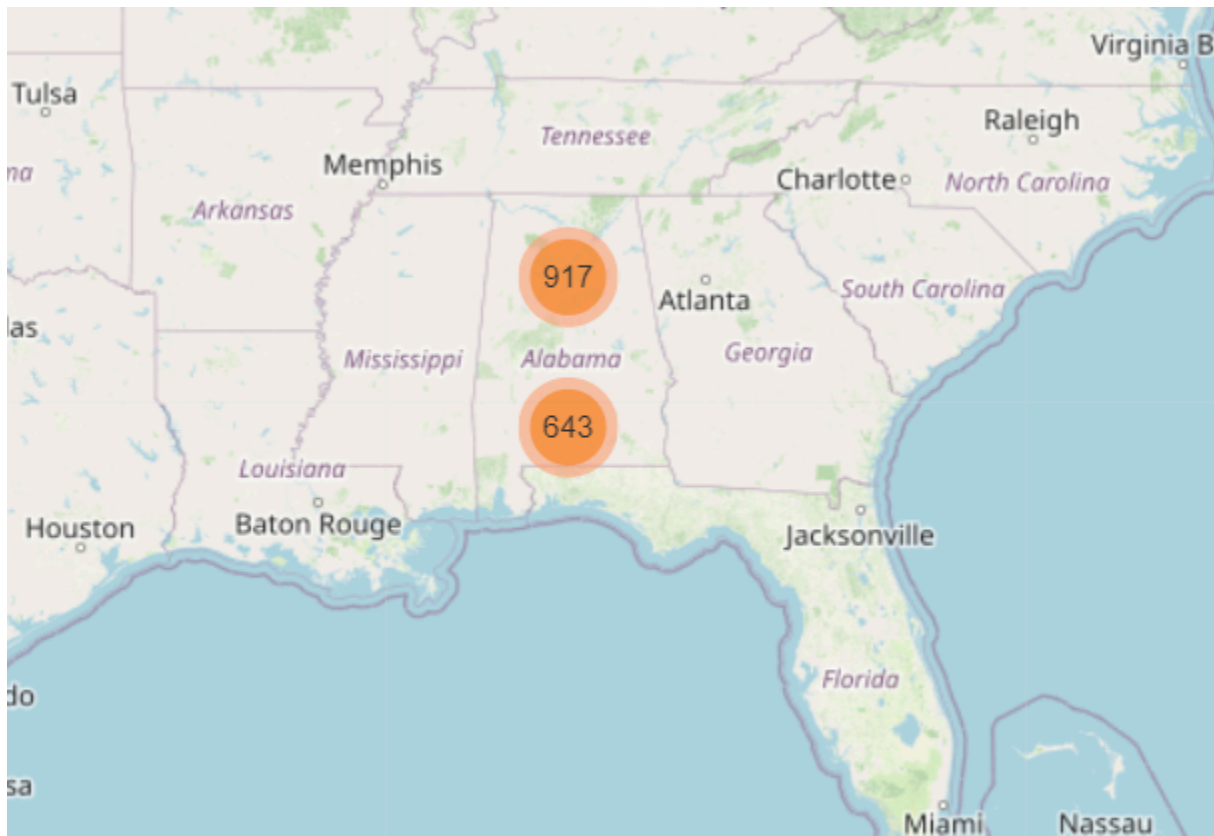
Our approach was to first make a grid across the map and group the latitudes and longitudes into their respective block on the grid to simplify stuff, we made sure that the blocks of the grid are not too big, but big enough to gather all lat longs in a specific amount of a road. We then threw in the data into this grid from past years, we first tested this mechanism on Alabama to see if we successfully made a system that shows reasonable hotspots:



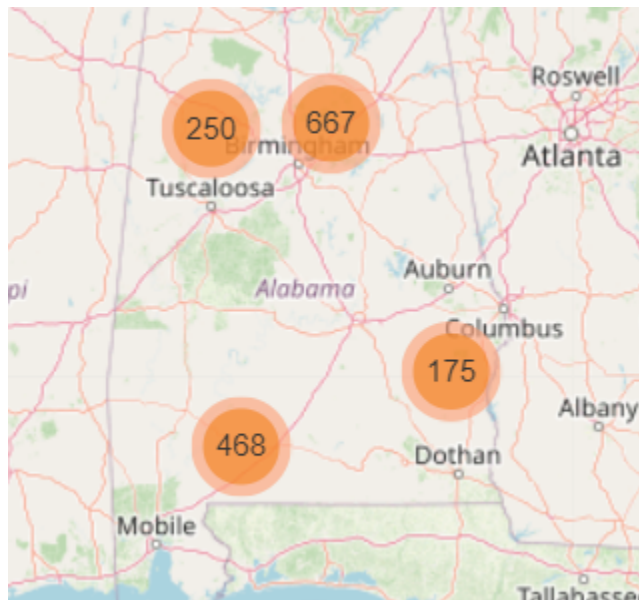
We succeeded in that, as you can see in the graph above, we made a very reasonable distribution of the entire area and made the boxes of the grid as containers that keep the frequency of the accidents happened in that specific area. We use boxes (groups of lat longs) as landmarks instead of using a single point for reference.

After this successful demonstration, we mapped the data on an actual map:

(zoomed out view)

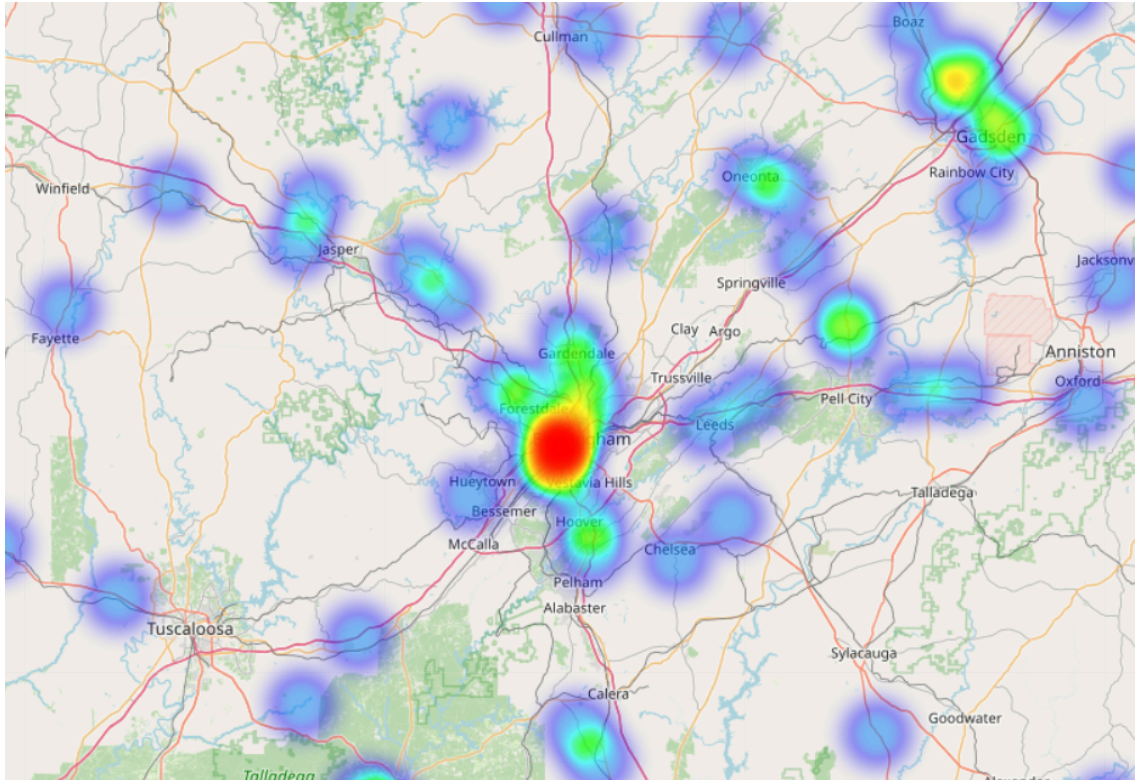


Let's zoom in steadily:

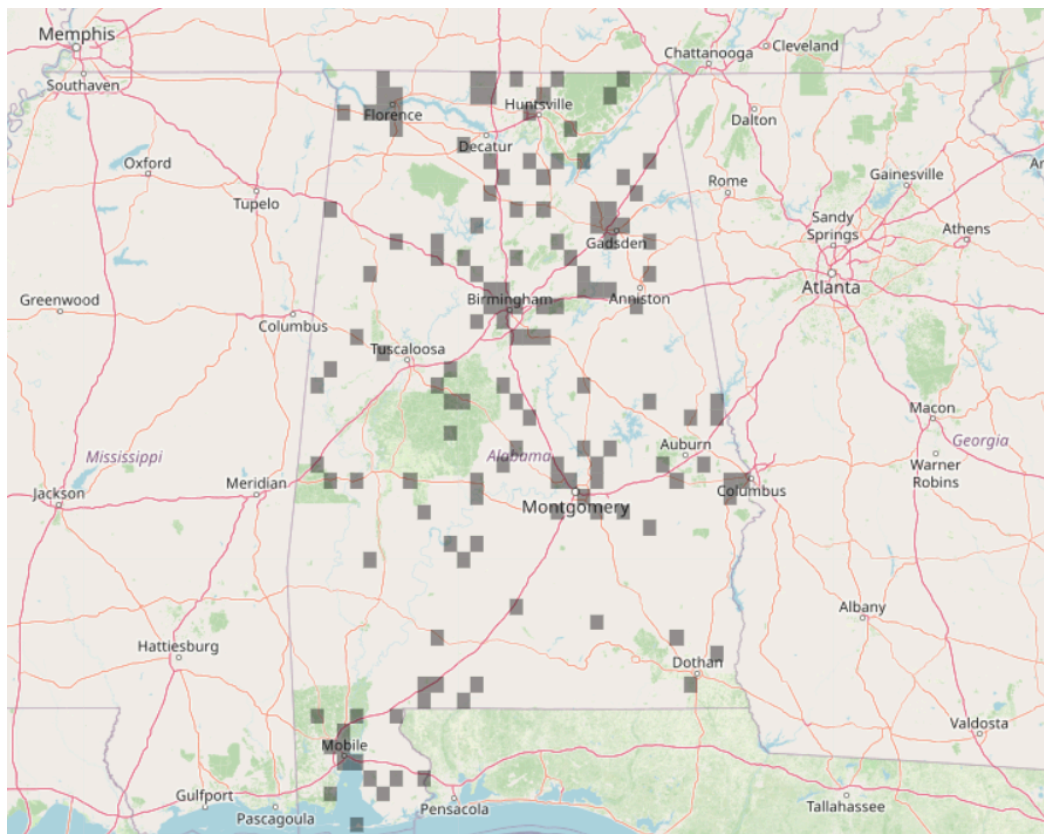


As we zoom in, these blobs break into further detailed blobs indicating high risk areas. These are just for data analysis and user viewing purposes if needed.

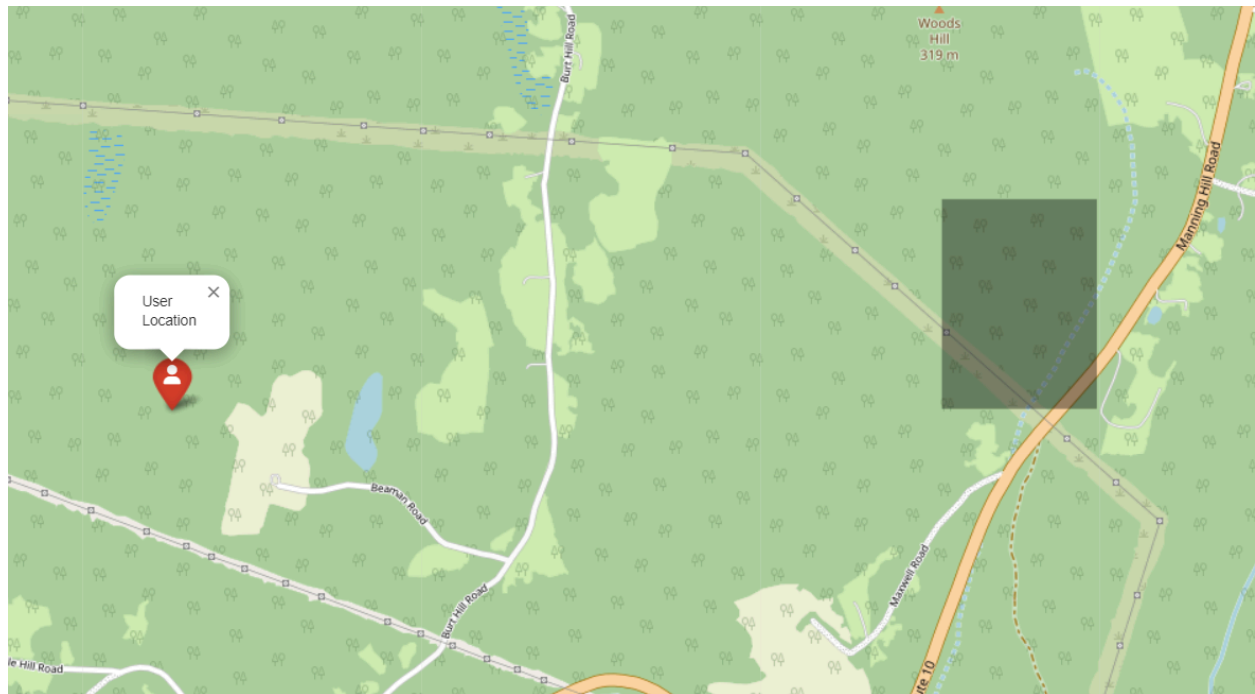
Then we have heatmaps, that show hot spots very clearly:



To understand the “Grid Method” that we have employed, this is a good representation, the dark blocks are areas where accidents occurred:



To explain how our alert system would work, the image below is a good reference, we keep tracking user location which is indicated by the red mark on the map, and as soon as the user gets into a close vicinity of the “grey box” which is an area marked indicating higher accident rates, the user gets an alert:



Why is this system better than the one deployed before?

1. We don't count every accident as it making that area dangerous, only after a “box” in the “grid” surpasses the 50% threshold either w.r.t its immediate surroundings or closest recorded accidents, it will be treated as a dangerous zone, otherwise if a box has low number of accidents below 50% threshold w.r.t its surroundings, it won't show up on the map. This mitigates the unreasonable fear that marking all accidents could produce and provide reasonable alerts that are actually useful.
2. The box and grid method is capable of marking “areas” as dangerous rather than a single point as dangerous, as accidents may have found to be happened on multiple lat longs in the same area and that would not help us identify if an “Area” is something we should alert the user about, the box method marks areas and the size of this can be adjusted as needed.

What are we working on now?

Now, we are building the FAST API endpoint to take in user information and return if an area is alert worthy or not.